



Lab 08: Implement Cloud Security Fundamentals - Challenge Lab

Challenge Lab | Cloud Security Fundamentals | GSP342



This is a Challenge Lab — no step-by-step instructions provided.
Skills from Labs 01-07 were applied independently to complete all tasks.

Overview

Configured a secure Kubernetes Engine private cluster for the Orca team using a dedicated service account with least privilege. Created a custom IAM role for Cloud Storage access, bound IAM roles to the service account, deployed a private GKE cluster, and tested access from the jump host.

Tasks Completed



Task 1 - Create a custom security role



Pass



Task 2 - Create a service account



Pass



Task 3 - Bind IAM security roles to the service account



Pass



Task 4 - Create and configure a private Kubernetes Engine cluster



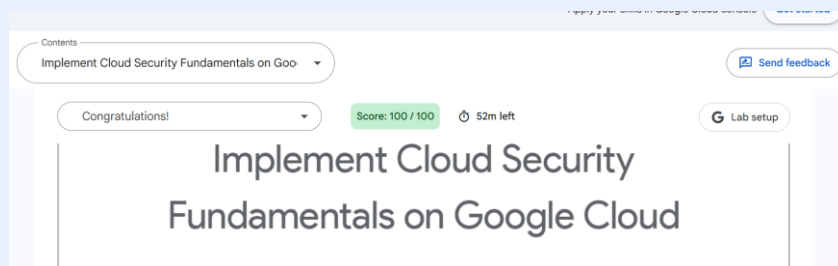
Pass



Task 5 - Deploy an application to the private cluster



Pass



Task 1 - Create Custom Security Role

Created a custom IAM role with Cloud Storage permissions for the Orca team's service account.

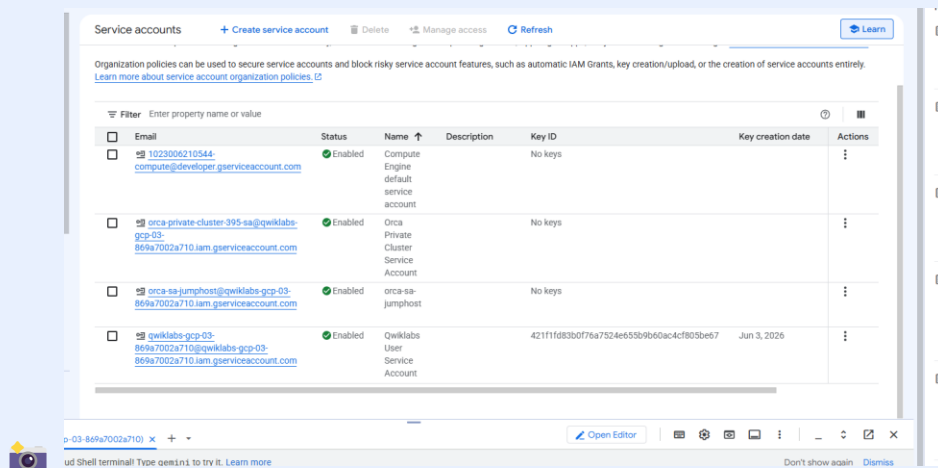
- Permissions included: storage.buckets.get, storage.objects.get, storage.objects.list, storage.objects.update, storage.objects.create

```
gcloud iam roles create [CUSTOM_ROLE] --project=$DEVSHHELL_PROJECT_ID --  
title="[ROLE_TITLE]" --  
permissions=storage.buckets.get,storage.objects.get,storage.objects.list,storage.ob  
jects.update,storage.objects.create
```

```
student_01_a49ccdca1a92@cloudshell:~ (qwiklabs-gcp-03-869a7002a710) $ gcloud iam roles create orca_storage_editor_321 \  
--project=qwiklabs-gcp-03-869a7002a710 \  
--title="Orca Storage Editor" \  
--permissions=storage.buckets.get,storage.objects.get,storage.objects.list,storage.objects.update,storage.objects.create  
Created role [orca_storage_editor_321].  
etag: Bw2ImW4xmi4=  
includedPermissions:  
- storage.buckets.get  
- storage.objects.create  
- storage.objects.get  
- storage.objects.list  
- storage.objects.update  
name: projects/qwiklabs-gcp-03-869a7002a710/roles/orca_storage_editor_321  
stage: ALPHA  
title: Orca Storage Editor  
student_01_a49ccdca1a92@cloudshell:~ (qwiklabs-gcp-03-869a7002a710) $
```

Task 2 - Create Service Account

```
gcloud iam service-accounts create [SA_NAME] --display-name "[SA_DISPLAY_NAME]"
```



The screenshot shows the Google Cloud IAM Service Accounts page. At the top, there are tabs for 'Service accounts', '+ Create service account', 'Delete', 'Manage access', and 'Refresh'. Below the tabs, there is a filter bar and a table of service accounts. The table has columns for 'Email', 'Status', 'Name', 'Description', 'Key ID', 'Key creation date', and 'Actions'. The table lists four service accounts: 'compute@developer.gserviceaccount.com', 'orca-private-cluster-395-sa@qwiklabs-gcp-03-869a7002a710.iam.gserviceaccount.com', 'orca-sa-jumphost@qwiklabs-gcp-03-869a7002a710.iam.gserviceaccount.com', and 'qwiklabs-gcp-03-869a7002a710@qwiklabs-gcp-03-869a7002a710.iam.gserviceaccount.com'. The first three are 'Enabled' and have 'No keys'. The last one is also 'Enabled' and has a key ID '421f1f583b0f76a7324e55b9b60ac4cf805be67' and a key creation date of 'Jun 3, 2026'.

Email	Status	Name	Description	Key ID	Key creation date	Actions
compute@developer.gserviceaccount.com	Enabled	Compute Engine default service account		No keys		
orca-private-cluster-395-sa@qwiklabs-gcp-03-869a7002a710.iam.gserviceaccount.com	Enabled	Orca Private Cluster Service Account		No keys		
orca-sa-jumphost@qwiklabs-gcp-03-869a7002a710.iam.gserviceaccount.com	Enabled	orca-sa-jumphost		No keys		
qwiklabs-gcp-03-869a7002a710@qwiklabs-gcp-03-869a7002a710.iam.gserviceaccount.com	Enabled	Qwiklabs User Service Account		421f1f583b0f76a7324e55b9b60ac4cf805be67	Jun 3, 2026	

Task 3 - Bind IAM Roles to Service Account

Bound the three required GKE roles plus the custom storage role to the service account.

```
gcloud projects add-iam-policy-binding $DEVSHHELL_PROJECT_ID --
member=serviceAccount:[SA_EMAIL] --role=roles/monitoring.viewer
gcloud projects add-iam-policy-binding $DEVSHHELL_PROJECT_ID --
member=serviceAccount:[SA_EMAIL] --role=roles/monitoring.metricWriter
gcloud projects add-iam-policy-binding $DEVSHHELL_PROJECT_ID --
member=serviceAccount:[SA_EMAIL] --role=roles/logging.logWriter
gcloud projects add-iam-policy-binding $DEVSHHELL_PROJECT_ID --
member=serviceAccount:[SA_EMAIL] --
role=projects/$DEVSHHELL_PROJECT_ID/roles/[CUSTOM_ROLE]
```

The screenshot shows the Google Cloud IAM console interface. At the top, it identifies the 'Service account: 1131417348039531354 / Access'. Below this, there are tabs for 'Details', 'Permissions', 'Keys', 'Metrics', and 'Logs', with 'Principals with access' being the active tab. A section titled 'Principals with access to this service account' provides instructions on granting access. A checkbox for 'Include Google-provided role grants' is present. Below this, there are links to 'View by principals' and 'View by roles'. A 'Filter' input field is also visible. The main part of the screenshot is a table listing the principals with access to the service account.

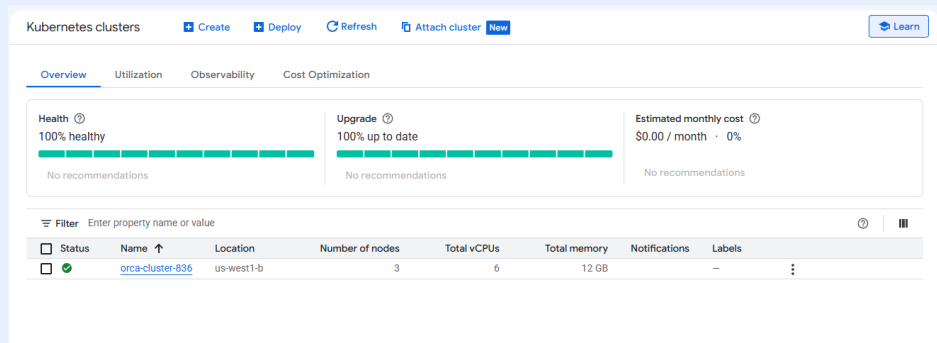
Type	Principal	Name	Role	Inheritance
Service account	1023006210544-compute@developer.gserviceaccount.com	Compute Engine default service account	Editor	qwiklabs-gcp-03-869a7002a710
Service account	admiral@qwiklabs-services-prod.iam.gserviceaccount.com		Owner	qwiklabs-gcp-03-869a7002a710
Service account	orca-sa-jumphost@qwiklabs-gcp-03-869a7002a710.iam.gserviceaccount.com	orca-sa-jumphost	Owner	qwiklabs-gcp-03-869a7002a710
Service account	qwiklabs-gcp-03-869a7002a710@qwiklabs-gcp-03-869a7002a710.iam.gserviceaccount.com	Qwiklabs User Service Account	Owner	qwiklabs-gcp-03-869a7002a710
User	student-01-a49ccda1a92@qwiklabs.net	student cbb0e199	Editor	qwiklabs-gcp-03-869a7002a710
			Owner	qwiklabs-gcp-03-869a7002a710
			Service Account Admin	qwiklabs-gcp-03-869a7002a710
			Viewer	qwiklabs-gcp-03-869a7002a710

Task 4 - Create Private Kubernetes Engine Cluster

Created a private GKE cluster in the orca-build-subnet with the dedicated service account and master authorized network set to the jumhost's internal IP.

```
gcloud container clusters create [CLUSTER_NAME] --zone=[ZONE] --service-
account=[SA_EMAIL] --network=orca-build-vpc --subnetwork=orca-build-subnet --
enable-master-authorized-networks --enable-ip-alias --enable-private-nodes --
enable-private-endpoint --master-ipv4-cidr=172.16.0.16/28
```

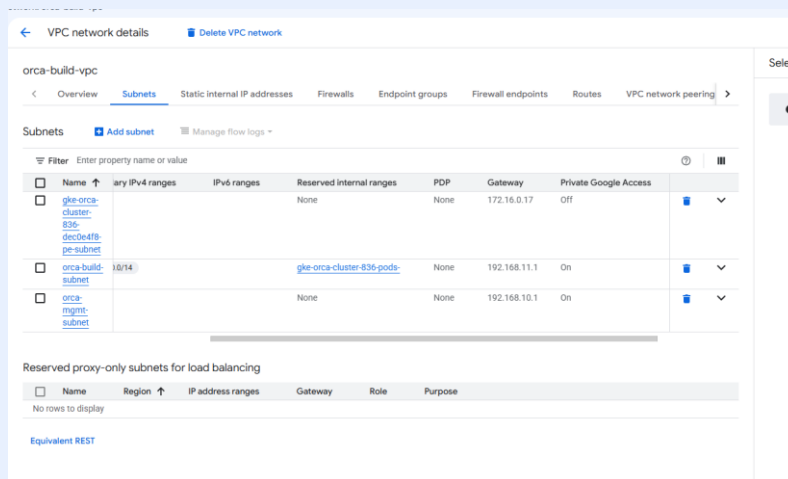
```
gcloud container clusters update [CLUSTER_NAME] --enable-master-authorized-networks
--master-authorized-networks=[JUMPHOST_IP]/32 --zone=[ZONE]
```



Task 5 - Deploy Application to Private Cluster

SSH'd into orca-jumphost, installed kubectl and gke-gcloud-auth-plugin, connected to the cluster using internal IP, and deployed the hello-server application.

```
sudo apt-get install google-cloud-sdk-gke-gcloud-auth-plugin
gcloud container clusters get-credentials [CLUSTER_NAME] --internal-ip --
project=[PROJECT_ID] --zone=[ZONE]
kubectl create deployment hello-server --image=gcr.io/google-samples/hello-app:1.0
```



What I Learned

- Custom IAM roles enforce least privilege by including only needed permissions
- GKE clusters need monitoring and logging roles at minimum for proper operation

- Private clusters with private endpoints can only be accessed from within the VPC
- Master authorized networks restrict kubectl access to specific IP ranges
- A jump host inside the VPC is required to manage a fully private cluster
- The gke-gcloud-auth-plugin is required for kubectl to authenticate with GKE

Challenges & How I Solved Them

- [Add any issues you ran into and how you solved them]

Reflection

This challenge lab combined custom IAM roles, service accounts, and private Kubernetes clusters into one real-world security scenario. Completing all 5 tasks proves practical knowledge of cloud security fundamentals on Google Cloud.

Resources

GCP GKE Security Docs: <https://cloud.google.com/kubernetes-engine/docs/how-to/hardening-your-cluster>

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